Math 432/532: Introduction to Topology II

Solutions to HW #7

1. Let $f : S^n \to S^n$ be a map with $\deg(f) \neq 1$. Show that there is a point $x \in S^n$ with $f(x) = -x$.

Solution: Suppose not, and let $g = -\mathrm{id}_{S^n} \circ f : S^n \to S^n$. Then g has no fixed points, so its degree is $(-1)^{n+1}$. But we have $\deg(g) = \deg(-\mathrm{id}_{S^n}) \cdot \deg(f) = (-1)^{n+1}\deg(f)$, which contradicts the hypothesis that $\deg(f) \neq 1$.

2. Let $f : S^n \to S^n$ be a map with n even. Show that there is a point $x \in S^n$ with $f(f(x)) = x$.

Solution: If f has a fixed point, we're done, so we may assume that it does not. If f has no fixed points, then f is homotopic to the antipodal map via the homotopy

$$F(x,t) := \frac{(1-t)f(x) - tx}{||(1-t)f(x) - tx||}.$$

In this case, $f \circ f$ is homotopic to the identity map via the homotopy

$$G(x,t) := F(F(x,t),t).$$

Finally, we proved in class that if n is even and g is homotopic to the identity, then g has a fixed point. Applying this to $g = f \circ f$, we get the desired statement.

Another solution: If f has a fixed point, we're done, so we may assume that it does not. If f has no fixed points, then $\deg(f) = (-1)^{n+1}$, so $\deg(f \circ f) = (-1)^{n+1} \cdot (-1)^{n+1} = 1 \neq (-1)^{n+1}$. But we proved that any map from S^n to itself with degree not equal to $(-1)^{n+1}$ has a fixed point.

Yet another: We have $\deg(f \circ f) = \deg(f)^2 \geq 0$, therefore $\Lambda_{f \circ f} = 1 + (-1)^n \deg(f \circ f) = 1 + \deg(f \circ f) > 0$. Since the Lefschetz number is nonzero, $f \circ f$ must have a fixed point.

3. Show that, if n is odd, $-\mathrm{id}_{S_n} \sim \mathrm{id}_{S_n}$. (Hint: Try doing it when $n = 1$ first.)

Solution: Suppose $n = 2m - 1$, and write $x = (x_1, \ldots, x_{2m}) \in S^n \subset \mathbb{R}^{2m}$. Let

$$F(x,t) = (\cos(\pi t)x_1 - \sin(\pi t)x_2, \sin(\pi t)x_1 + \cos(\pi t)x_2, \ldots, \cos(\pi t)x_{2m-1} - \sin(\pi t)x_{2m}, \sin(\pi t)x_{2m-1} + \cos(\pi t)x_{2m}).$$

This is the desired homotopy.

4. Let K be a nonempty connected simplicial complex, and suppose that $f : |K| \to |K|$ is homotopic to a constant map. Show that there exists $x \in |K|$ with $f(x) = x$.

Solution: A constant map from K to itself induced the identity map on $H_0(K)$ and the zero map on $H_i(K)$ for all $i > 0$, and therefore has Lefschetz number 1. It follows that f has Lefschetz number 1, and therefore has a fixed point.

5. Suppose that $f : |K| \to |K|$ is simplicial and that the set $S := \{x \mid f(x) = x\}$ is finite. You may assume that every element of S is a vertex of K^1, which we stated but did not prove earlier in the course. More precisely, $S = \{\hat{A} \mid A \in K, f(A) = A\}$.

(i) Show that, if $f(A) = A$, then there is no proper face $B \subsetneq A$ such that $f(B) = B$.

Solution: If there were, then the entire edge of K^1 connecting $\hat{B}$ to $\hat{A}$ would be fixed by f, which would contradict the fact that S is finite.

(ii) Suppose that $f(A) = A$. Show that one can order the vertices of A $(v_0, \dots, v_i)$ such that $f(v_j) = v_{j+1}$ for all $0 \le j \le i-1$ and $f(v_i) = v_0$.

Solution: Pick any vertex of A and call it v_0. Let $v_1 = f(v_0)$, $v_2 = f(v_1)$, and so on. We just need to show that we cannot have $f(v_j) = v_0$ for any $j < i$. Indeed, if we did, then the simplex spanned by $v_0, \dots, v_j$ would be fixed by f, but part (i) says that this cannot happen.

(iii) Let $\sigma = (v_0, \dots, v_i)$ be the orientation of A that you constructed in part (ii). Show that $f_i(\sigma) = (-1)^i \sigma$.

Solution: We know that $f_i(\sigma) = (v_1, \dots, v_i, v_0)$, and this is related to σ by i simple transpositions.

(iv) Show that $\Lambda_f = |S|$.

Solution: We proved in class that

$$\Lambda_f = \sum_{i=0}^{\dim K} (-1)^i \operatorname{tr}\left(f_i : C_i(K) \to C_i(K)\right).$$

Let $A_1, \dots, A_r$ be the i-simplices of K, with $A_1, \dots, A_s$ fixed by f. and for each $1 \le j \le r$, let σ_j be some choice of orientation for A_j, so that $\sigma_1, \dots, \sigma_j$ form a basis for $C_i(K)$. We showed in part (iii) that $f(\sigma_j) = (-1)^i \sigma_j$ for all $j \le s$, so the trace of f_i is equal to $(-1)^i s$. Multiplying by $(-1)^i$ and taking the sum over all i, we see that each element of S contributes 1 to Λ_f.